Model Testing and Refinement Report

Week 7 · Data Science track: model testing, error analysis and refinement

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Bottom line

The Week 6 candidate was tested properly for the first time and does not hold up. It reproduces exactly (AUC 0.505), but on a locked hold-out its AUC is 0.479 (95% CI 0.423-0.535); a label-shuffle test gives p = 0.42; trained on January-August it scores 0.467 on September-December; and a train AUC of 0.75 against a validation AUC of 0.51 shows heavy over-fitting.

Refinement fixed what could be fixed: over-fitting (gap 0.24 to 0.06), unreliable features, a threshold that flagged 95% of appointments, the missing release gate, and silent failures at the model interface (10/15 to 15/15 contract tests). It could not create signal: the refined hold-out AUC is 0.455 (CI 0.398-0.513) and its permutation p is 0.73. The release gate fails for both versions.

Two positive controls (Age 65+ and the rule-based high-risk flag) are recovered, so the harness can detect signal when it exists. The null result is therefore a property of the current data, not of a weak test. Recommendation: do not use the model for any operational decision; repair the data and re-run this suite unchanged.

Week 6 to Week 7 transition

Item	Answer
Main Week 6 output	Data Science package (Data Quality, EDA, Statistical Analysis, Predictive Analytics reports and executed notebook)
Most important result	Task 3 admission-proxy classifier (target: Emergency visit with Severe diagnosis). Best model gradient boosting, grouped 5-fold CV AUC 0.505 (Week 6 reported 0.505)
Main issue carried over	Evidence was cross-validation only; threshold tuned on the same predictions; Week 6 itself questioned the facility, department and BMI fields that stayed in the model; no admissions or attendance outcome table
Component tested	Candidate classifier, its preprocessing pipeline, and its model interface
Relevant tracks	Data Science (owner), ML Engineering (interface), Data Analytics (finding validation), Project Management (issues, readiness)
What was tested	Reproducibility, locked hold-out, stability, permutation test, over-fitting, time-based validity, calibration, operating point, errors, segments, feature reliability, leakage, target scan with positive controls
Expected improvement / validation	Replace optimistic CV-only evidence with a locked-test verdict; remove unreliable features; fix the threshold; add a release gate; deliver a tested interface to ML Engineering

Testing readiness

Acceptance criteria were fixed before any test was run. M = mandatory for the model to be called suitable; S = supporting quality criteria.

ID	Type	Criterion	Pass rule
AC1	S	Reproducibility	Re-running the Week 6 protocol reproduces the reported AUC (0.505) within +/-0.005
AC2	M	Discrimination on locked test set	Hold-out AUC >= 0.60 and the lower 95% bootstrap bound > 0.50
AC3	M	Operational value	Precision in the top-20% highest-scored appointments (lift) >= 1.30 x prevalence, lower bound of lift > 1.0
AC4	S	Stability	At least 90% of 10 repeated grouped-CV runs have AUC > 0.55
AC5	M	Signal vs chance	Label-permutation test p < 0.05
AC6	S	Calibration	Calibration slope within 0.8-1.2 and Brier score better than the prevalence-only forecast
AC7	S	Over-fitting	Train-minus-validation AUC gap <= 0.05
AC8	S	Time-based validity	AUC >= 0.55 when trained on Jan-Aug and tested on Sep-Dec
AC9	M	Meaningful improvement over baseline	Paired-bootstrap delta-AUC vs baseline >= 0.05 with the 95% lower bound > 0

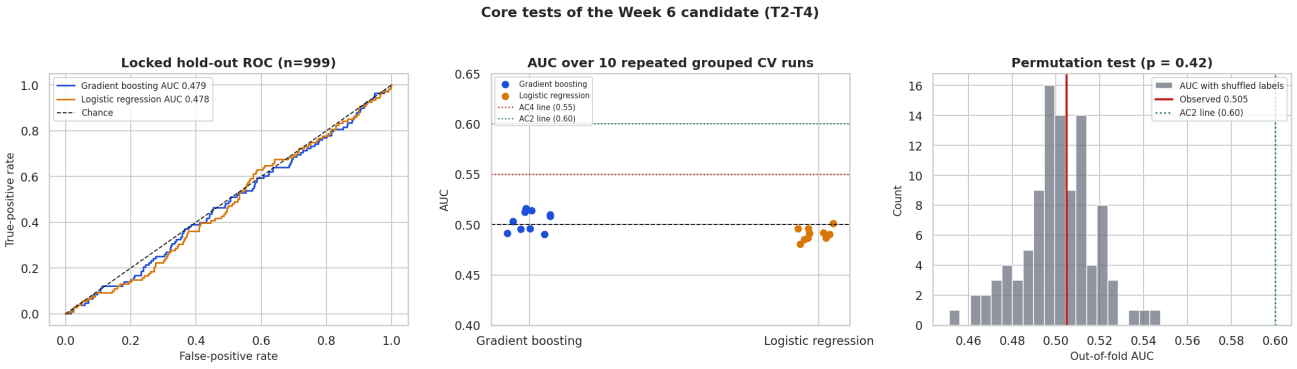
ID	Type	Criterion	Pass rule
AC10	S	Segment parity	Recall and flag-rate gap between sex, age-band and province groups <= 10 percentage points (groups with n >= 100)

Test results on the Week 6 model as built

Test	Expected	Actual	Result
T1 re-run Week 6 CV protocol	AUC 0.505; confusion [[221,4235],[21,523]]	AUC 0.5049; confusion reproduced exactly	PASS
T2 locked 20% hold-out	AUC >= 0.60, CI above 0.5	AUC 0.479 (95% CI 0.423-0.535)	FAIL
T3 10 repeated grouped CV	90% of runs AUC>0.55	0% of runs AUC>0.55; mean 0.504	FAIL
T4 label permutation (100)	p < 0.05	p = 0.416	FAIL
T5 train vs validation AUC	Gap <= 0.05	train 0.746 - validation 0.506 = 0.240	FAIL
T6 train Jan-Aug, test Sep-Dec	AUC >= 0.55	AUC 0.467 (CI 0.427-0.513)	FAIL
T7 slope and Brier	Slope 0.8-1.2	slope 0.10 (CI -0.25-0.45); Brier 0.0975 vs prevalence 0.0970	FAIL
T8 policy comparison	Sensible flag rate, lift > 1.3	F1 threshold flags 95%, lift 1.01x; top-20% lift 1.06x	FAIL
T9 FP vs TN, FN vs TP at top-20%	Identifiable, fixable error groups	flagging driven by FacilityID, BMI_clean, Age (Holm p<0.001) but positive rate flagged 0.115 vs unflagged 0.107	FAIL
T10 sex/age/province/facility	Gap <= 10 pp	max recall gap 20.5 pp; max flag-rate gap 22.5 pp	FAIL
T11 audit, ablation, leakage	All features reliable and informative	4 unreliable; ablation deltas within noise; no leakage found; importance>2SD: 0/14	FAIL (I2)
T12 positive controls + target scan	Controls recovered; other targets reveal any signal	Controls recovered: True; detected non-control outcomes: ['Returning patient (PriorVisits>0)'] (returning-patient: month artifact, vanishes when month removed)	PASS
Paired bootstrap delta-AUC	delta >= 0.05, CI > 0	delta-AUC 0.006 (95% CI -0.028 to 0.039)	FAIL

Locked test set: 999 appointments (108 positives), patient-grouped, used once per model version. The decision threshold was chosen by nested CV on training data only.

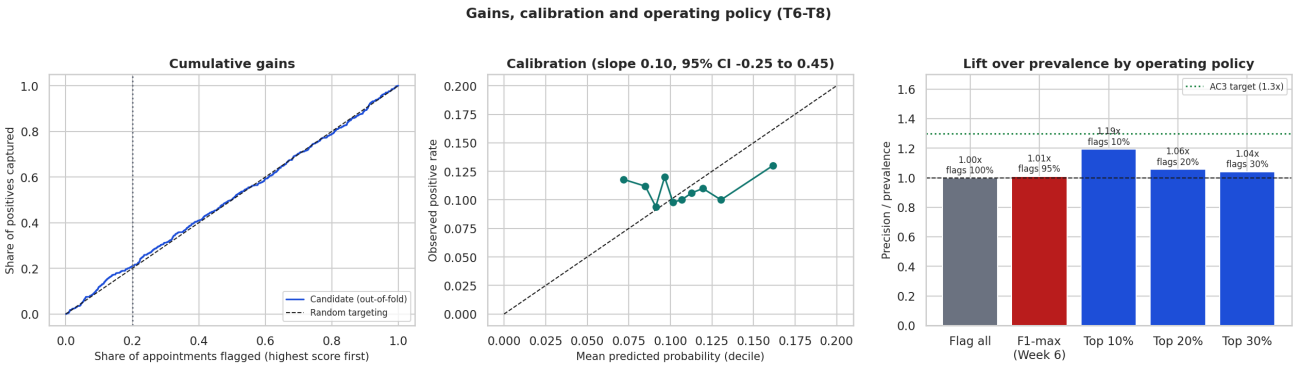
Figure: Core tests of the Week 6 candidate (T2-T4)



Interpretation: On a locked hold-out of 999 appointments the candidate scores AUC 0.479 (95% CI 0.423-0.535); across 10 repeated CV runs the AUC ranges 0.490-0.516, and the observed AUC sits inside the cloud of AUCs produced by shuffled labels (p = 0.42). The Week 6 result is reproducible, but it is a null result.

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Figure: Gains, calibration and operating policy (T6-T8)



Interpretation: Targeting by score captures positives at almost exactly the random-targeting rate (gain curve on the diagonal), predicted probabilities span only 0.04-0.28, and the Week 6 F1-maximising threshold flags 95% of appointments with lift 1.01x: an operating point that is indistinguishable from flagging everyone.

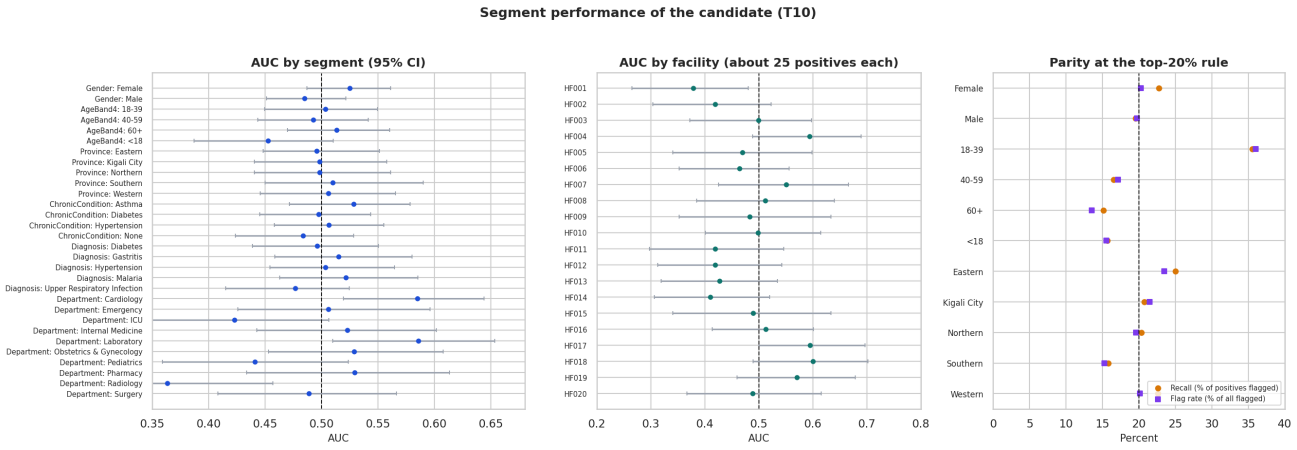
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Error analysis and segments

At the top-20% flag rule the model's flagged and unflagged appointments differ clearly on FacilityID, BMI_clean, Age, which shows the score follows those fields. But the positive rate is 0.115 among flagged and 0.107 among unflagged (prevalence 0.109), so the choice buys no accuracy. This is noise-fitting rather than a fixable error pattern, and it matches the over-fitting found in T5. Because facility and BMI are among the fields the score follows, they were removed in refinement.

Segment AUCs sit between 0.36 and 0.60; only 4 of 50 intervals exclude 0.5, which chance alone produces. Parity fails (AC10): the share flagged differs by up to 22 percentage points between age bands. Because accuracy is the same in every group, this is arbitrary rather than risk-based, but it would matter the moment a real signal exists (issue I12).

Figure: Segment performance of the candidate (T10)



Interpretation: AUC is near 0.5 in every segment (0.36-0.60); only 4 of 50 intervals exclude 0.5, about what chance alone gives. But who gets flagged is uneven: the flag rate ranges 14%-36% across age bands, because the score follows noise in facility, BMI and age. Recall tracks the flag rate and precision stays at prevalence in every group, so the unequal flagging buys no accuracy.

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Feature review and leakage

Feature	Source	Week 6 evidence	Grade
Age	Patients.DateOfBirth	Consistent with DOB; but uniform 0-90, no age gradient in any outcome (Week 6 EDA)	Valid, uninformative
Gender	Patients.Gender	Valid; ~50/50; names do not match sex	Valid, uninformative
HasChronic / ChronicCondition	Patients.ChronicCondition	75% prevalence in every age band, 1,005 children with hypertension (Week 6 EDA)	Questionable coding

Feature	Source	Week 6 evidence	Grade
Diagnosis	Diagnoses	5 diagnoses, each ~20%; assigned at consultation, so not known at booking	Valid, timing caveat
PriorVisits	Appointments (derived)	Counts earlier visits only (verified in section 0.1); but extract holds 22% of patients and 8% of billing history	Valid, truncated history
MonthNum / WeekdayNum	Appointments.Date	874 appointments dated after the 24 Oct 2026 analysis date carry diagnoses and bills	Valid, future-dated rows
DoctorYears	Doctors.YearsExperience	Doctor's home department differs from appointment department in 99.4% of rows	Questionable link
DeptWorkloadIndex	Derived from departments	Computed on all 5,000 appointments (including future test rows); staffing 4-9 per department	Derived on full data
DepartmentName	Departments	Appointment facility differs from the department's owning facility in 94.9% of rows (Critical, Week 6)	Unreliable
FacilityID	Appointments.FacilityID	Same 94.9% disagreement; 20 one-hot columns for ~25 positives each	Unreliable
BMI_clean	Patients.BMI	Set to missing for under-18s (19% missing); adult-range height/weight recorded for 2,594 children	Unreliable
ProvinceName	Geography	Patient to village to province chain is clean; districts are placeholders	Valid, uninformative

Feature group dropped	AUC (3 x 5-fold CV)	Change
None (all 14 features)	0.503 ± 0.005	+0.0000
Unreliable (Facility, Department, BMI)	0.507 ± 0.010	+0.0042
Geography (Province)	0.501 ± 0.006	-0.0021
Clinical (Diagnosis, Chronic)	0.511 ± 0.007	+0.0082
Demographic (Age, Gender)	0.491 ± 0.008	-0.0122
Utilisation (Prior visits, Month, Weekday)	0.507 ± 0.002	+0.0041
Provider (Doctor yrs, Dept workload)	0.502 ± 0.005	-0.0010

Dropping any group changes AUC by less than about 0.012, within noise; dropping the unreliable group changes it by +0.0042. No feature has permutation importance beyond 2 SD on the locked test set (largest 0.0158). Leakage audit: Target columns absent from features: PASS; PriorVisits counts only strictly earlier visits: PASS; DeptWorkloadIndex recomputed inside each training fold: PASS; Diagnosis available when the prediction is made: FLAG.

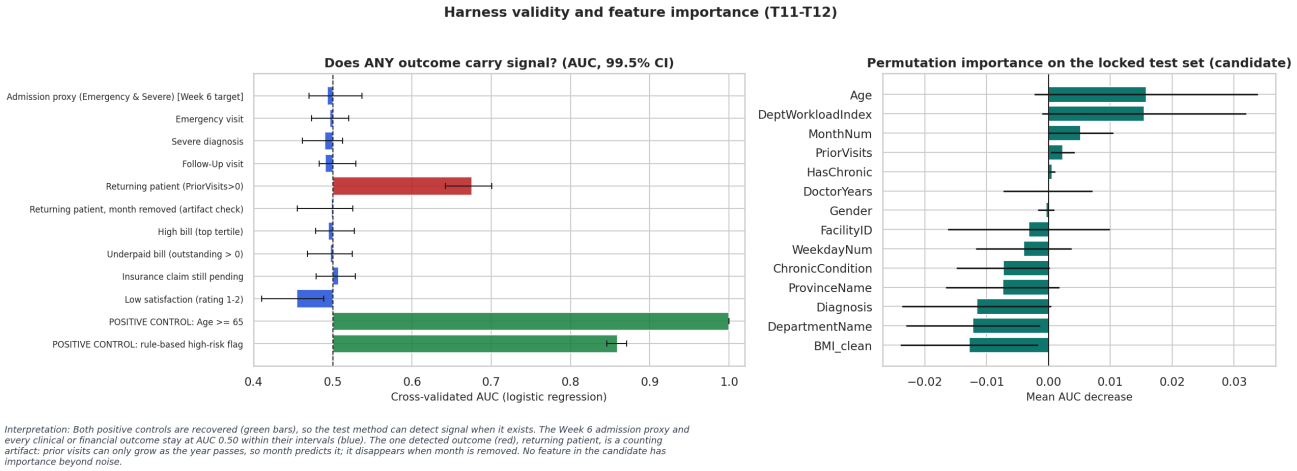
Data sensitivity scenario	n	Positives	AUC
All 5,000 appointments (as built)	5,000	544	0.503 ± 0.005
Exclude 874 future-dated appointments	4,126	453	0.504 ± 0.008
Adults only (BMI valid)	4,030	442	0.499 ± 0.008
First visit per patient only	4,443	488	0.504 ± 0.011

Target scan and positive controls

Target	n	Prevalence	AUC	99.5% CI	Signal
Admission proxy (Emergency & Severe) [Week 6 target]	5,000	0.11	0.493	0.470-0.537	none
Emergency visit	5,000	0.33	0.496	0.473-0.520	none
Severe diagnosis	5,000	0.32	0.490	0.462-0.512	none
Follow-Up visit	5,000	0.34	0.491	0.483-0.529	none
Returning patient (PriorVisits>0)	5,000	0.11	0.676	0.642-0.701	DETECTED
Returning patient, month removed (artifact check)	5,000	0.11	0.498	0.455-0.526	none
High bill (top tertile)	5,000	0.33	0.495	0.479-0.527	none
Underpaid bill (outstanding > 0)	5,000	0.20	0.497	0.468-0.525	none
Insurance claim still pending	3,521	0.50	0.507	0.479-0.529	none
Low satisfaction (rating 1-2)	1,756	0.41	0.455	0.410-0.489	none
POSITIVE CONTROL: Age >= 65	5,000	0.29	1.000	1.000-1.000	DETECTED
POSITIVE CONTROL: rule-based high-risk flag	5,000	0.41	0.859	0.846-0.871	DETECTED

The returning-patient target is predicted only through the month variable (prior visits can only accumulate during the year); when month is removed the signal disappears. Positive controls are recovered, which validates the method. Data Analytics also notes the target's positive rate (10.88%) is almost exactly what two independent labels would give (10.71%), which fits the chance-level result.

Figure: Harness validity and feature importance (T11-T12)



Both positive controls are recovered (green bars), so the test method can detect signal when it exists. The Week 6 admission proxy and every clinical or financial outcome stay at AUC 0.50 within their intervals (blue). The one detected outcome (red), returning patient, is a counting artifact: prior visits can only grow as the year passes, so month predicts it; it disappears when month is removed. No feature in the candidate has importance beyond noise.

Baseline versus candidate

The Week 5 baseline model is not in the supplied files, so the prevalence baseline and the Week 6 logistic regression serve as references. The minimum meaningful improvement was fixed in advance: delta-AUC of at least 0.05 with a 95% lower bound above zero (paired bootstrap).

Evaluation	Comparison	Delta AUC	95% CI	p	Meaningful?
Out-of-fold, n=5,000	Candidate vs Prevalence baseline	+0.006	-0.028 to +0.039	0.74	NO
Out-of-fold, n=5,000	Candidate vs Logistic regression	+0.010	-0.017 to +0.037	0.44	NO
Locked hold-out, n=999	Candidate vs Prevalence baseline	-0.021	-0.077 to +0.032	0.47	NO
Locked hold-out, n=999	Candidate vs Logistic regression	+0.001	-0.053 to +0.056	0.93	NO

The improvement over baseline is not meaningful in any comparison; the differences are well inside sampling noise.

Refinement and re-testing

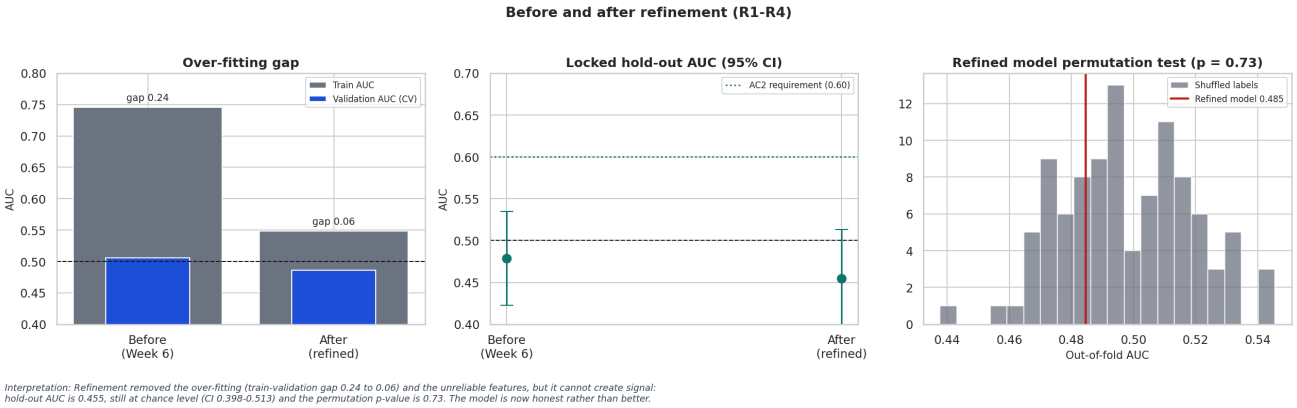
#	Issue found	Evidence	Refinement
I1	Severe over-fitting	Train AUC 0.75 vs validation 0.51	R2 regularised models, chosen by nested CV
I2	Unreliable features in the model	94.9% facility/department mismatch; BMI invalid for minors	R1 remove FacilityID, DepartmentName, BMI_clean, DeptWorkloadIndex
I3	Degenerate operating point	F1 threshold flags 95% of appointments (lift 1.0x)	R3 flag-rate budget (top 20%), reported with lift
I4	No release rule	A no-signal model could be deployed	R4 executable release gate AC1-AC10
I5	No predictive signal anywhere	T4, T6, T11, T12	Not fixable by modelling; escalated as a data issue

Candidates were compared by repeated grouped CV on the training 80% only (simplest model within one standard error of the best); the locked test set was then scored once for the chosen model.

Candidate	Features	CV AUC	SE	Train AUC	Gap
A. As built (GB, 14 features)	14	0.506	0.0010	0.751	0.244
B. Shallow GB, 10 reliable features	10	0.491	0.0008	0.612	0.118
C. Ridge logistic, 10 reliable features	10	0.506	0.0043	0.563	0.054
D. Ridge logistic, 5 core features	5	0.505	0.0048	0.545	0.033

Selected: C. Ridge logistic, 10 reliable features.

Figure: Before and after refinement (R1-R4)



Interpretation: Refinement removed the over-fitting (train-validation gap 0.24 to 0.06) and the unreliable features, but it cannot create signal: hold-out AUC is 0.455, still at chance level (CI 0.398-0.513) and the permutation p-value is 0.73. The model is now honest rather than better.

Refinement removed the over-fitting (train-validation gap 0.24 to 0.06) and the unreliable features, but it cannot create signal: hold-out AUC is 0.455, still at chance level (CI 0.398-0.513) and the permutation p-value is 0.73. The model is now honest rather than better.

Version	Features	Train AUC	CV AUC	Gap	Hold-out AUC (CI)	Time-split AUC	Top-20% lift	Perm. p	Mandatory met
Before: Week 6 candidate	14	0.746	0.506	0.240	0.479 (0.423-0.535)	0.467	0.78	0.42	no
After: C. Ridge logistic, 10 reliable features	10	0.549	0.486	0.063	0.455 (0.398-0.513)	0.484	0.85	0.73	no

Release gate, side by side

ID	Type	Criterion	Week 6 candidate	Result	Refined model	Result
AC1	S	Reproducibility	max |diff| = 0.0008	PASS	n/a (new model)	n/a
AC2	M	Hold-out discrimination	AUC 0.479 (95% CI 0.423-0.535)	FAIL	AUC 0.455 (95% CI 0.398-0.513)	FAIL
AC3	M	Operational lift (top 20%)	lift 0.78x (95% CI 0.44-1.15)	FAIL	lift 0.85x (95% CI 0.54-1.18)	FAIL
AC4	S	Stability (10 runs)	0% of runs AUC>0.55; mean 0.504	FAIL	0% of runs AUC>0.55; mean 0.480	FAIL
AC5	M	Permutation test	p = 0.416	FAIL	p = 0.733	FAIL
AC6	S	Calibration	slope 0.10 (CI -0.25-0.45); Brier 0.0975 vs prevalence 0.0970	FAIL	slope -0.27 (CI -0.84-0.30); Brier 0.0973 vs prevalence 0.0970	FAIL
AC7	S	Over-fitting gap	train 0.746 - validation 0.506 = 0.240	FAIL	train 0.549 - validation 0.486 = 0.063	FAIL
AC8	S	Time-based AUC	AUC 0.467 (CI 0.427-0.513)	FAIL	AUC 0.484 (CI 0.440-0.532)	FAIL
AC9	M	Improvement vs baseline	delta-AUC 0.006 (95% CI -0.028 to 0.039)	FAIL	delta-AUC -0.015 (95% CI -0.049 to 0.018)	FAIL
AC10	S	Segment parity	max recall gap 20.5 pp; max flag-rate gap 22.5 pp	FAIL	max recall gap 29.3 pp; max flag-rate gap 35.6 pp	FAIL

Refinement removed over-fitting and unreliable inputs but not the absence of signal. Calibration (AC6) is judged on predicted probabilities versus observed rates, so a model with no signal can fail it even when the ranking is unchanged.

HC-POD cross-track testing

A. Data Science to ML Engineering: interface contract. Test, finding, action, retest. The same 15 interface tests were run against the Week 6 pipeline as an ML pipeline would call it, then against the refined scorer, then in a fresh process from the hand-off folder. No ML Engineering teammate was available, so this is a Data Science-side test and is not claimed as a collaboration.

ID	Scenario	Expected	Week 6 pipeline	Refined scorer
C1	Valid batch of 5	Scores in [0,1], no null	PASS	PASS
C2	Single row	Scored	PASS	PASS
C3	Missing numeric values	Scored (imputed)	PASS	PASS

ID	Scenario	Expected	Week 6 pipeline	Refined scorer
C4	Missing categorical value	Scored, flagged	PASS	PASS
C5	Unseen category ('Zanzibar' in a categorical column)	Scored AND visible in status	FAIL	PASS
C6	Required column missing	Clear error naming the column	PASS	PASS
C7	Extra unexpected column	Ignored, scores unchanged	PASS	PASS
C8	Columns in different order	Same scores	PASS	PASS
C9	Numbers sent as text ('45')	Coerced, same scores	PASS	PASS
C10	Non-numeric value ('abc') in Age	Bad row rejected, rest of batch scored	FAIL	PASS
C11	Impossible values (Age 250, PriorVisits -3)	Bad rows rejected, rest scored	FAIL	PASS
C12	Wrong text case ('female','MALARIA')	Normalised, same scores	FAIL	PASS
C13	Empty input	Empty output, no crash	FAIL	PASS
C14	Repeat call determinism	Identical output	PASS	PASS
C15	Duplicate row index	Scored	PASS	PASS

Result: 10/15 passed before, 15/15 after; fresh-process run exit code 0; save-and-reload scores identical: True. The Week 6 pipeline scored impossible values silently, failed whole batches on one bad row, hid unseen categories and returned opaque errors. The scorer rejects bad rows with a reason, scores the rest, normalises text, reports a status per row and labels the model 'no predictive signal'.

B. Data Analytics to Data Science: validating findings that drive modelling decisions. Week 6 findings were recomputed from the raw project files.

Finding	Week 6	Recomputed	Status	Effect on modelling
Appointment facility differs from department's owning facility	94.9% (4,744 rows)	94.9% (4,744 rows)	CONFIRMED	FacilityID and DepartmentName removed (R1); ablation shows no AUC loss
Bills whose appointment is missing	55,000	55,000	CONFIRMED	Appointment-level modelling limited to 5,000 linked rows
Diagnoses whose appointment is missing	55,000	55,000	CONFIRMED	Same limit; more modelling data cannot be obtained until keys are repaired
Appointments dated after 24 Oct 2026	874	874	CONFIRMED	Sensitivity test T11d: excluding them does not change AUC
Bill Status label carries no payment information	Paid 90.0% / Partial 90.1% / Pending 90.2% paid (p=0.643)	Paid 90.0% / Partial 90.1% / Pending 90.2% paid (p=0.643)	CONFIRMED	BillStatus never used as a feature (satisfaction model excluded it from conclusions)
Follow-Up visits with no earlier visit in the extract (NEW)	not reported in Week 6	88.0%	NEW	NEW finding for Data Analytics: VisitType or the extract's history is incomplete; PriorVisits is a truncated-history feature

Independent cross-check received from Data Analytics (Olabisi Amusan)

Olabisi's validated-findings summary was compared line by line with my own recomputation (rows 1-7). I then asked Olabisi to check three things against the source files, and the results (rows 8-11) are in her reply. Evidence: Olabisi_Amusan_Short_Test_Summary.docx and My_finding.docx.

#	Data Analytics finding (Olabisi Amusan)	My Week 7 result	Agreement	Effect on the model / project
1	5,000 appointments vs 60,000 bills, diagnoses and payments; only 5,000 link via AppointmentID; linked records resemble unlinked ones	55,000 orphan bills and diagnoses recomputed from raw files	AGREE	Appointment-level modelling stays at 5,000 rows. Their 'linked resembles unlinked' point is new: it supports treating the 5,000 as a fair sample (taken as reported, not retested by me)
2	FacilityID differs from the department's facility in 4,744 of 5,000 appointments; same conflict in staff records (1,187 of 1,250)	4,744 (94.9%) recomputed. I removed FacilityID and DepartmentName instead of substituting the department's facility	AGREE on count; different handling	Staff-record conflict is new: DoctorYears (doctor link) stays graded 'questionable'. Both approaches avoid the conflicting field
3	No attended / missed / cancelled field, so a no-show target cannot be built	Same conclusion (issue I8)	AGREE	Confirmed by two tracks; escalated to Project Management

#	Data Analytics finding (Olabisi Amusan)	My Week 7 result	Agreement	Effect on the model / project
4	Weak signal: severity independent of diagnosis (p=0.19), department (0.20), visit type (0.91); satisfaction by facility p=0.55, department 0.88, visit type 0.68; age vs visits 0.012	Diagnosis vs severity p=0.214; satisfaction by facility p=0.795, by department 0.878; age vs billing/admission proxy rho near 0	AGREE on conclusion; some p-values differ	Same finding: nothing is significant. P-values differ for facility satisfaction (0.55 vs 0.80) and diagnosis-severity (0.19 vs 0.21), probably from different tests or samples; not reconciled, conclusion unaffected
5	Classes close to balanced (visit types about a third each; claims 50/50), so accuracy is weak	Week 6 EDA: same splits	AGREE	Already handled: model judged on AUC, PR-AUC, lift and permutation test, not accuracy
6	Repeat visits rare: 4,443 patients, 5,000 appointments (average 1.13)	22.2% of patients appear; PriorVisits mean 0.12	AGREE	PriorVisits carries little information, consistent with its near-zero importance
7	Phone numbers corrupted to scientific notation; every patient has an insurance ID	Same Week 6 data-quality findings	AGREE	Neither field used in the model
8	Follow-up check: 88.0% of Follow-Up visits have no earlier visit (88.1% with strictly earlier dates); same for other visit types (89.3%); only 510 patients have more than one appointment	Recomputed 88.0% (Follow-Up) and 89.3% (others)	AGREE	Follow-Up label does not reflect visit history; VisitType already excluded from the model; PriorVisits stays a weak feature
9	DoctorYears: doctor join is clean (100% of DoctorIDs found, 1-25 years) but doctor department matches the appointment department in only 0.6% and facility in 5.1%, about random	Week 6 data-quality finding: 5.14% facility match, 4,968 of 5,000 department mismatches	AGREE	DoctorYears kept as a doctor-level attribute only, never read as department or facility staffing
10	Target: one diagnosis per appointment (join is safe); 544 of 5,000 positives (10.88%); 10.71% expected if Emergency and Severe were independent	Recomputed: observed 10.88% vs expected 10.71%	AGREE	New insight: the proxy behaves like two unrelated labels combined, which explains the chance-level AUC
11	HasChronic check: 5,071 zeros and 14,929 ones ('None' is read as missing by default in pandas)	Verified: pipeline reads with keep_default_na=False; 1,263 'None' kept, 0 missing	AGREE	No change needed

Cross-track activity record

Field	A. ML Engineering interface test	B. Data Analytics validation
Track	ML Engineering (Data Science side only, no counterpart)	Data Analytics (Olabisi Amusan), two-way
Project dependency	Data Science to ML Engineering: model inputs and outputs	Data Analytics to Data Science: findings that influence features
Component tested	Model artefact and interface	Five Week 6 data-quality findings
Information received	None: no ML Engineering counterpart was available in the pod	'Week 7 Validated Findings' summary (7 findings), then a written reply to my three checks (follow-up finding, DoctorYears, target and HasChronic)
Information provided	Scorer, schema, model, 15 contract tests, model status	Target, features, before/after metrics and baseline; three questions to check against the source data
Testing activity completed	15 contract tests, before and after (DS side complete)	Both sides tested the same findings against the source files: my recomputation, their independent validation, and their checks of my three questions
Issue / finding	Silent failures at the interface (10 of 15 passed)	All overlapping findings agree. Analytics confirmed my 88% follow-up finding (88.0%; 89.3% for other visit types), and added: staff-record facility conflict (1,187 of 1,250), doctor department/facility match only 0.6%/5.1% (about random), and that the target behaves like two unrelated labels (10.88% observed vs 10.71% expected if independent)
Action taken	Validated wrapper, schema, normalisation, row-level status	No model change needed: unreliable fields were already removed, DoctorYears kept as a doctor-level attribute only, HasChronic verified correct. Limitation text and issue I9 updated

Field	A. ML Engineering interface test	B. Data Analytics validation
Retest result	15/15 pass; fresh-process exit 0. Not run by a counterpart; prepared for Week 8 hand-off	All three of Analytics' checks agree with my values; metrics unchanged (hold-out AUC 0.455, CI 0.398-0.513), so there is no before/after change to report
What changed	Model now cannot be consumed without visible status and validation	Follow-Up label and DoctorYears limits documented; the chance-level result now has an explanation
Evidence	handoff_ml_engineering folder; contract_tests_before/after.csv	Notebook section 7.1; Olabisi_Amusan_Short_Test_Summary.docx; My_finding.docx
Improvement to HealthConnect	Safer integration for Week 8	Modelling decisions traceable to validated findings

Suitability, limitations and next steps

Question	Answer	Evidence
Is the model accurate enough to use?	No	Hold-out AUC 0.479 before and 0.455 after refinement; permutation p = 0.42 / 0.73; time-split AUC 0.47
Are its strengths and weaknesses understood?	Yes	Fully characterised: no signal in any feature (T11), in any alternative target (T12), in any segment (T10); test harness proven by positive controls
Is it safe to hand to ML Engineering?	Yes, as a tested interface only	15/15 contract tests pass (was 10/15); model carries an explicit 'no predictive signal' status
Is it suitable for the HealthConnect use case?	No	Mandatory criteria AC2, AC3, AC5 and AC9 fail for both versions; gate result: FAIL. Using it would assign arbitrary risk scores to patients
What is the right use of the work now?	Re-usable test harness	The same suite re-runs unchanged when admissions or attendance outcomes arrive; refined feature set and release gate are the starting point

Limitations

Only 5,000 appointments (544 positives; 108 in the locked test set) are available, so intervals are wide. The target is a proxy: no admission, length-of-stay, referral or attendance outcome exists in the supplied tables. The Week 5 baseline is not among the supplied files. The ML Engineering interface test was done on the Data Science side only because no counterpart was available; the Data Analytics validation was completed with Olabisi Amusan. Diagnosis is known only after consultation, so a booking-time model could not use it.

Issue log

ID	Issue	Severity	Owner	Status	Action
I1	Over-fitting in the Week 6 candidate (train AUC 0.75 vs validation 0.51)	High	Data Science	MITIGATED	Regularised model on reliable features (R1, R2): gap 0.06 (AC7 limit 0.05 still narrowly missed)
I2	Unreliable features (facility/department 94.9% mismatch, BMI for minors) inside the model	High	Data Science / Data Analytics	RESOLVED	Removed; ablation shows no performance loss (T11b)
I3	F1-maximising threshold flags 95% of appointments (lift 1.0x)	Medium	Data Science	RESOLVED	Top-20% flag-rate budget with lift reporting (R3)
I4	No release gate: a no-signal model could be deployed	High	Data Science / Project Management	RESOLVED	Executable gate AC1-AC10 (R4); result recorded in schema.json
I5	Silent failures at the model interface (impossible values, mis-cased text, whole-batch failures)	High	Data Science / ML Engineering	RESOLVED (DS side); no ML Engineering counterpart available	Scorer wrapper; contract tests 10/15 -> 15/15
I6	No predictive signal in any available feature or alternative target (data looks simulated)	Critical	Data Analytics / Project Management / data owner	OPEN	Escalated: needs real outcome data; cannot be fixed by modelling
I7	No Admissions table: admission is modelled through a proxy (Emergency & Severe)	Critical	Project Management / data owner	OPEN	Obtain Admissions, Referrals, Laboratory and Prescriptions tables

ID	Issue	Severity	Owner	Status	Action
I8	No attendance / no-show field in the supplied tables, so the Week 7 brief's attendance questions cannot be tested on this dataset	Critical	Project Management / Data Analytics	OPEN	Confirm which dataset the pod uses for attendance; harness is ready to re-run
I9	Only 5,000 of 60,000 transactions link to an appointment; 22% of patients appear; Follow-Up visits without earlier visits (88%, confirmed by Data Analytics: the Follow-Up label does not reflect history)	High	Data Analytics / data owner	OPEN	Repair appointment keys; re-extract full history
I10	Diagnosis is recorded at consultation, so a booking-time model cannot use it	Medium	Data Science / ML Engineering	OPEN	Decide prediction time-point before Week 8 integration
I11	Week 5 baseline model not available in the supplied files	Low	Data Science	OPEN	Prevalence baseline and logistic regression used as reference points
I12	Uneven flagging by age band (flag rate gap 36 pp, AC10 fails before and after) although accuracy is the same in every group	Medium	Data Science	OPEN	Moot while the model is not deployed; re-run parity checks once a real signal exists

Decision log

ID	Decision	Reason
D1	Acceptance criteria fixed before testing; test set locked and used once per model version	Prevents tuning to the test data
D2	Model chosen on training data only; simplest model within 1 SE of the best (C. Ridge logistic, 10 reliable features)	Parsimony; no evidence of benefit from complexity
D3	Remove FacilityID, DepartmentName, BMI_clean, DeptWorkloadIndex	Unreliable links (T11a); no AUC loss (T11b)
D4	Operating point = flag-rate budget (top 20%), reported with lift	F1 threshold degenerates to flag-all when there is no signal
D5	Do NOT release the model for any operational use; release only the tested interface and harness	Gate FAIL on mandatory criteria
D6	Do not search for a more 'successful' target after the scan; the Week 6 proxy stays the reference	Avoid p-hacking: 9 targets scanned with Bonferroni intervals, none detected
D7	Report the Data Analytics validation as a two-way cross-track test and the ML Engineering interface test as Data Science side only (no counterpart available)	Only activities that actually happened are claimed

Actions before and during Week 8

ID	Action	Owner	When
W8-1	Obtain Admissions / Referrals / Attendance outcome data and re-run notebook sections 2-5 unchanged	Project Management / data owner	Before final integration
W8-2	ML Engineering (when assigned) runs hc_contract_tests.py inside the pipeline and returns the result; fix any mismatch	ML Engineering / Project Management	Early Week 8
W8-3	Agree the prediction time-point (booking vs after consultation) and the allowed feature list	Data Science + ML Engineering (once a counterpart is assigned)	Early Week 8
W8-4	Data Analytics confirms the follow-up/prior-visit finding and repairs appointment keys	Data Analytics	Week 8
W8-5	Present the HealthConnect model as 'validated null result + reusable test harness', not as a prediction tool	All tracks	Final presentation
W8-6	Add fairness checks by sex, age band and province once a signal exists	Data Science	After real data